# Customer Churn Prediction: A Comparative Analysis Using Bayesian and Frequentist Approaches

**STAT-I 529: INTRODUCTION TO BAYESIAN STATISTICS**

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**Abstract**

This project aims to predict customer churn and analyze contributing factors using both Bayesian and frequentist methods. By leveraging a comprehensive dataset of customer information, we develop models to identify potential churners and provide insights into the uncertainty of predictions. The analysis employs various statistical techniques, including exploratory data analysis, hypothesis testing, and Bayesian modeling, to gain a deeper understanding of customer behavior and improve retention strategies.

## 1 Introduction

Customer churn, the loss of customers to competitors, significantly impacts profitability and customer retention. Predicting churn is essential for devising effective retention strategies.

This study compares Bayesian and frequentist approaches to analyze customer data and predict churn. By examining factors such as demographics and service usage, the study aims to provide actionable insights. Bayesian methods, with their ability to quantify uncertainty, enhance model reliability and support data-driven decision-making for proactive churn mitigation.

## 2 Statement of Bayesian Approach for Customer Churn Prediction

In this project, we utilize the Bayesian framework to predict the likelihood of customer churn (whether a customer will leave or stay with the company) based on customer demographics, service usage patterns, and other relevant features. The Bayesian approach incorporates uncertainty into churn predictions, aiding stakeholders in making informed decisions regarding customer retention strategies.
**Outcome:** The outcome variable, $Y_{\text{churn}}$, is binary, where:

$$Y_{\text{churn}} = \begin{cases} 1 & \text{if the customer churns (leaves the company),} \\ 0 & \text{if the customer stays.} \end{cases}$$

The probability of churn, $P(Y_{\text{churn}} = 1 \mid X)$, is modeled based on the input feature vector $X$ (e.g., customer demographics, service usage patterns, etc.).

## 3 Bayesian Logistic Regression

Bayesian Logistic Regression models the probability of a binary outcome $y_i \in \{0, 1\}$ given input features $\mathbf{x}_i$ and model parameters $\mathbf{w}$.

### 3.1 Prior

The prior distribution $P(\mathbf{w})$ reflects our beliefs about the model parameters before observing any data. Common choices include:

$$P(\mathbf{w}) \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad \text{or} \quad P(\mathbf{w}) \sim \text{Uniform}(-a, a)$$

where $\mathcal{N}(\mathbf{0}, \mathbf{I})$ is a normal distribution with mean 0 and identity covariance matrix.

### 3.2 Likelihood

The likelihood function describes how the observed data is generated given the parameters:

$$P(\mathbf{y}|\mathbf{X}, \mathbf{w}) = \prod_{i=1}^{N} \sigma(\mathbf{x}_i^T \mathbf{w})^{y_i} (1 - \sigma(\mathbf{x}_i^T \mathbf{w}))^{1-y_i}$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function. The likelihood captures how probable the observed outcomes $\mathbf{y}$ are given the model's parameters $\mathbf{w}$ and features $\mathbf{X}$.

### 3.3 Posterior

The posterior distribution combines the prior and likelihood, representing updated beliefs about the parameters after observing data:

$$P(\mathbf{w}|\mathbf{X}, \mathbf{y}) = \frac{P(\mathbf{y}|\mathbf{X}, \mathbf{w}) P(\mathbf{w})}{P(\mathbf{y}|\mathbf{X})}$$

where the marginal likelihood $P(\mathbf{y}|\mathbf{X})$ is:

$$P(\mathbf{y}|\mathbf{X}) = \int P(\mathbf{y}|\mathbf{X}, \mathbf{w}) P(\mathbf{w})\, d\mathbf{w}$$

The posterior reflects the updated knowledge about $\mathbf{w}$ after considering both prior information and observed data.

## 4 Methods

### 4.1 Data and Preprocessing

The analysis utilized the Telco Customer Churn dataset from Kaggle, originally published by IBM. The dataset contains customer records with attributes related to services, demographics, and billing. Key features include:

- **Churn:** Binary target variable indicating if a customer left within the last month.
- **Services:** Subscriptions to phone, internet, online security, backups, and streaming.
- **Account Information:** Tenure, contract type, payment method, monthly charges, and total charges.
- **Demographics:** Gender, partner, and dependent status.

Preprocessing steps included converting the `TotalCharges` column from string to numeric and removing rows with missing values. The `customerID` column was dropped as it held no predictive value. Categorical variables (e.g., `gender`, `Partner`, `InternetService`) were one-hot encoded, while the binary target variable (`Churn`) was encoded as 0 (No) and 1 (Yes). Numerical features (`tenure`, `MonthlyCharges`, `TotalCharges`) were normalized using `StandardScaler` from `scikit-learn`.

The cleaned dataset, containing 7,032 rows, was split into training (80%, 5,625 rows) and testing (20%, 1,409 rows) subsets to ensure rigorous evaluation on unseen data.

## 4.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) revealed key patterns and relationships in the dataset.

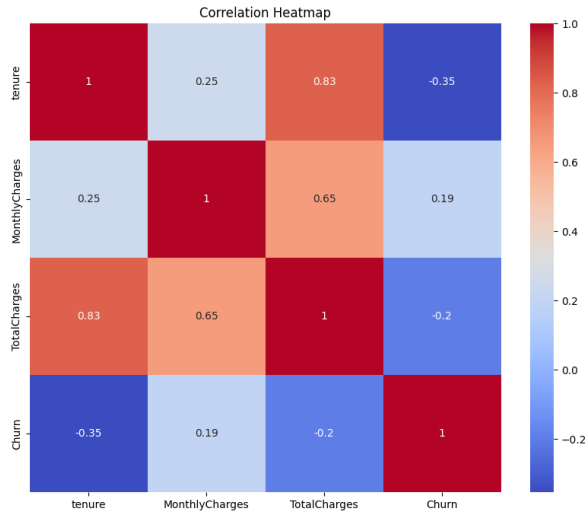



Figure 1: Correlation Matrix: Relationships among numerical features.

Figure 1 shows a strong positive correlation between `tenure` and `TotalCharges`, as longer tenure leads to higher total charges. Similarly, `MonthlyCharges` strongly correlates with `TotalCharges`, reflecting the cumulative nature of billing.

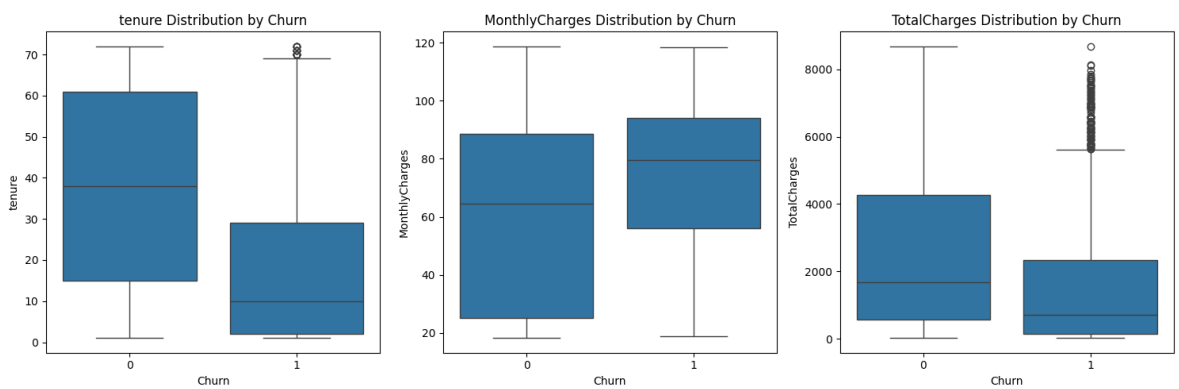



Figure 2: Distribution of Tenure, MonthlyCharges, and TotalCharges by Churn Status.

Figure 2 highlights feature distributions by churn status. Non-churned customers have longer tenures (median  37 months) compared to churned customers ( 10 months), with higher churn rates in early months. Churned customers pay higher monthly charges (median  80) than non-churned customers ( 65) and exhibit greater variability. Total charges are higher for non-churned customers due to longer tenure, though both groups display high variability and outliers.

## 4.3 Statistical Testing

Several statistical tests were conducted to explore relationships between features and customer churn.

1. A **Chi-Square test** found no significant association between **Gender** and **Churn** ($\chi^2 = 0.4847, p = 0.4865$), while a strong association was identified between **Contract type** and **Churn** ($\chi^2 = 1187.8376, p < 0.001$).
2. An **independent samples t-test** confirmed a significant difference in **Monthly Charges** between

churned and non-churned customers ($t = 18.5462, p < 0.001$).
3. **Levene's test** revealed a significant variance difference in **Tenure** between groups ($F = 245.8891, p < 0.001$).
4. A **two-proportion Z-test** indicated significant differences in **Online Security** proportions between churned and non-churned customers ($Z = -35.1874, p < 0.001$).
5. Finally, a **Pearson correlation test** showed a strong positive correlation between **Total Charges** and **Tenure** ($r = 0.8259, p < 0.001$).

## 4.4 Model Architecture

### 4.4.1 Prior Choice Selection and Weight Initialization

Priors for Bayesian logistic regression were selected using the Kolmogorov-Smirnov (KS) statistic to ensure they closely represent the observed data distributions, improving the model's ability to capture uncertainty. Binary variables were tested against `Bernoulli`, `Binomial`, `Poisson`, and `Uniform` distributions, while continuous variables were evaluated against `Normal`, `Uniform`, `Exponential`, `Log-Normal`, and `Gamma`. Figures 3 and 4 illustrate the prior selection process for `Tenure` and `Gender`, respectively. Final priors are summarized in Table 1.

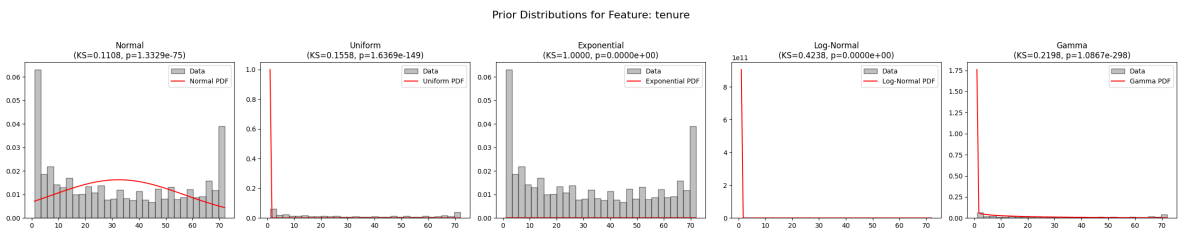



Figure 3: Choosing prior for "Tenure"

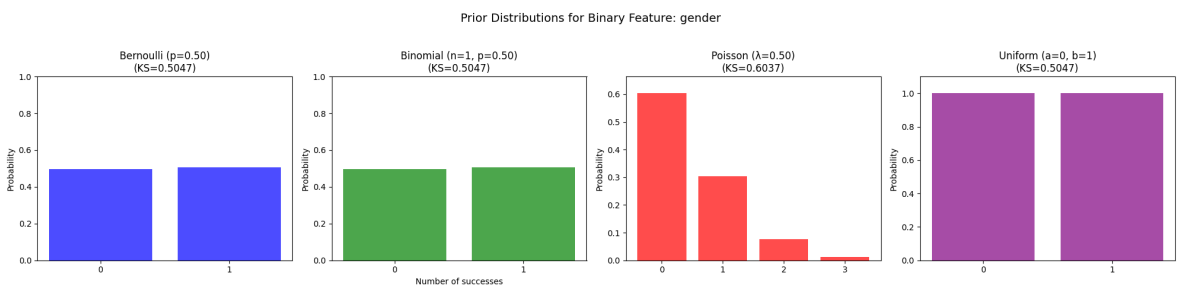



Figure 4: Choosing prior for "Gender"

Weight initialization was performed as follows:

- **Intercept:** Normal prior with $\mu = 0$ and $\sigma = 1$, $\texttt{intercept} \sim \mathcal{N}(0, 1)$.
- **Numerical Features:** Normal or Gamma priors: $\texttt{beta_tenure}, \texttt{beta_monthly_charges} \sim \mathcal{N}(0, 1)$, $\texttt{beta_total_charges} \sim \texttt{Gamma}(1, 1)$.
- **Binary Features:** Uniform priors bounded by $[-5, 5]$: $\texttt{beta_gender}, \texttt{beta_senior_citizen}, \ldots \sim \texttt{Uniform}(-5, 5)$.
- **Poisson-Modeled Features:** Features such as `PhoneService` and `PaperlessBilling` were also assigned Uniform priors bounded by $[-5, 5]$.

Table 1: Selected Priors for Features

| **Feature** | **Type** | **Prior Distribution** |
|---|---|---|
| Gender | Binary | Uniform |
| SeniorCitizen | Binary | Uniform |
| Partner | Binary | Uniform |
| Dependents | Binary | Uniform |
| PhoneService | Binary | Uniform |
| MultipleLines | Binary | Uniform |
| OnlineSecurity | Binary | Uniform |
| OnlineBackup | Binary | Uniform |
| DeviceProtection | Binary | Uniform |
| TechSupport | Binary | Uniform |
| StreamingTV | Binary | Uniform |
| StreamingMovies | Binary | Uniform |
| PaperlessBilling | Binary | Uniform |
| Churn | Binary | Uniform |
| IS_DSL | Binary | Uniform |
| IS_Fiber Optic | Binary | Uniform |
| Tenure | Continuous | Normal |
| MonthlyCharges | Continuous | Normal |
| TotalCharges | Continuous | Gamma |

#### 4.4.2 Sampling

The Bayesian logistic regression model was implemented using PyMC with the No-U-Turn Sampler (NUTS), an adaptive variant of Hamiltonian Monte Carlo (HMC) that optimizes hyperparameters during the warm-up phase. The model included an intercept and coefficients for numerical (`tenure`, `MonthlyCharges`, `TotalCharges`) and binary features (e.g., `gender`, `Partner`, `PhoneService`, etc.). Priors were assigned based on domain knowledge: Normal or Gamma priors for numerical features and Uniform priors for binary features.

The linear combination of features and coefficients was passed through the logistic function (`invlogit`) to model the probability of customer churn. The likelihood was specified as a Bernoulli distribution, with observed churn outcomes (`y_train`) as the response variable. Sampling was performed using 1,000 warm-up iterations for tuning, 1,000 posterior draws per chain, and 2 independent chains to assess convergence. The sampler was initialized using the `adapt_diag` method, and 3 CPU cores were employed for parallel processing.

This setup ensured efficient exploration of the posterior distribution while addressing issues such as poor convergence and biased estimates.

### 4.5 Posterior Analysis

#### 4.5.1 Posterior Summary

The posterior analysis shows the strength and direction of predictors' influence on churn, categorized into minimal, moderate, and strong effects.

**Minimal Effects ($|\textbf{mean}| < 0.2$):** Gender (-0.036), partner status (0.062), and streaming services (TV, movies  0.12) show negligible impact. The intercept (-0.955) indicates a negative baseline.

**Moderate Effects ($0.2 \leq |\textbf{mean}| < 0.5$):** Technical support (-0.645) and online security (-0.613) exhibit moderate negative effects. Total charges (0.605) have a moderate positive impact.

**Strong Effects ($|\textbf{mean}| \geq 0.5$):** Tenure (-1.655) has the strongest negative impact, significantly reducing churn. Phone service (-0.902) is also a strong negative predictor. Fiber optic service (0.782) shows the strongest positive association, increasing churn likelihood.

| | mean | sd | hdi_3% | hdi_97% | mcse_mean | mcse_sd | ess_bulk | ess_tail | r_hat |
|---|---|---|---|---|---|---|---|---|---|
| beta_dependents | -0.293 | 0.101 | -0.479 | -0.099 | 0.003 | 0.002 | 1618.0 | 1307.0 | 1.0 |
| beta_device_protection | -0.158 | 0.107 | -0.354 | 0.048 | 0.004 | 0.003 | 689.0 | 1142.0 | 1.0 |
| beta_gender | -0.036 | 0.075 | -0.179 | 0.093 | 0.002 | 0.002 | 1561.0 | 1408.0 | 1.0 |
| beta_is_dsl | 0.121 | 0.375 | -0.615 | 0.822 | 0.020 | 0.014 | 365.0 | 563.0 | 1.0 |
| beta_is_fiber_optic | 0.782 | 0.663 | -0.435 | 2.072 | 0.035 | 0.025 | 350.0 | 589.0 | 1.0 |
| beta_monthly_charges | 0.352 | 0.362 | -0.298 | 1.059 | 0.019 | 0.014 | 347.0 | 651.0 | 1.0 |
| beta_multiple_lines | 0.232 | 0.106 | 0.024 | 0.422 | 0.004 | 0.003 | 828.0 | 1021.0 | 1.0 |
| beta_online_backup | -0.191 | 0.101 | -0.386 | -0.004 | 0.004 | 0.003 | 770.0 | 1156.0 | 1.0 |
| beta_online_security | -0.613 | 0.106 | -0.818 | -0.421 | 0.004 | 0.003 | 818.0 | 1135.0 | 1.0 |
| beta_paperless_billing | 0.351 | 0.081 | 0.198 | 0.494 | 0.002 | 0.001 | 1491.0 | 1225.0 | 1.0 |
| beta_partner | 0.062 | 0.086 | -0.104 | 0.216 | 0.002 | 0.002 | 1494.0 | 1424.0 | 1.0 |
| beta_phone_service | -0.902 | 0.264 | -1.414 | -0.422 | 0.013 | 0.009 | 434.0 | 744.0 | 1.0 |
| beta_senior_citizen | 0.359 | 0.093 | 0.184 | 0.533 | 0.002 | 0.002 | 1739.0 | 1415.0 | 1.0 |
| beta_streaming_movies | 0.128 | 0.149 | -0.157 | 0.404 | 0.007 | 0.005 | 504.0 | 876.0 | 1.0 |
| beta_streaming_tv | 0.118 | 0.145 | -0.158 | 0.383 | 0.007 | 0.005 | 495.0 | 946.0 | 1.0 |
| beta_tech_support | -0.645 | 0.107 | -0.833 | -0.429 | 0.004 | 0.003 | 819.0 | 1376.0 | 1.0 |
| beta_tenure | -1.655 | 0.160 | -1.955 | -1.365 | 0.006 | 0.004 | 813.0 | 762.0 | 1.0 |
| beta_total_charges | 0.605 | 0.167 | 0.290 | 0.902 | 0.006 | 0.004 | 826.0 | 768.0 | 1.0 |
| intercept | -0.955 | 0.819 | -2.572 | 0.553 | 0.045 | 0.032 | 336.0 | 589.0 | 1.0 |

Figure 5: Posterior Summary of Model Parameters

## 4.6 Posterior Predictive Checks and Odds Ratio

The analysis reveals several key associations with customer behavior. Fiber optic service has the strongest positive impact (mean = 0.75), though with considerable uncertainty. Monthly charges (mean = 0.37) and senior citizen status (mean = 0.36) also show positive effects. Negative associations include phone service (mean = -0.91), tech support (mean = -0.55), and dependents (mean = -0.31). Notable positive effects are found for multiple lines (mean = 0.23), paperless billing (mean = 0.35), and tenure (mean = 0.17), all with reliable HDIs not crossing zero. Gender has a minimal effect (mean = -0.037), and streaming services show small positive effects, with HDIs crossing zero.

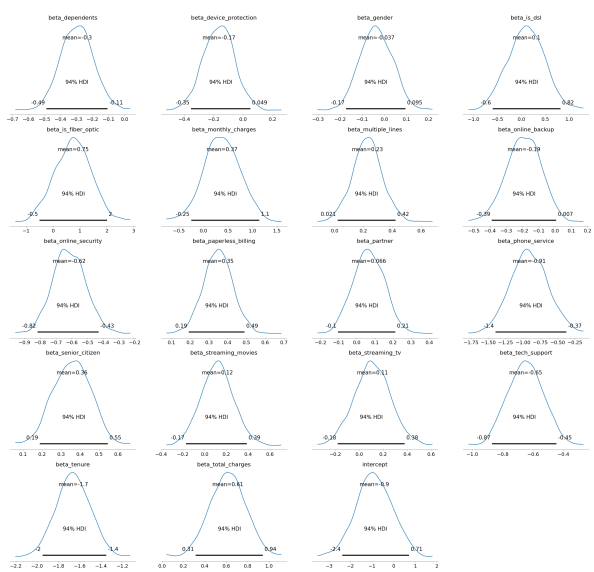



Figure 6: Trace Plot of Model Parameters

The trace plot shows the posterior predictive check, with the observed data (red line) centered around

0.3. The scattered vertical lines represent the distribution of predicted means, suggesting: - Model predictions range from 0 to 0.8. - The wide spread indicates uncertainty. - The observed value falls within the predicted range, implying a reasonable model fit.

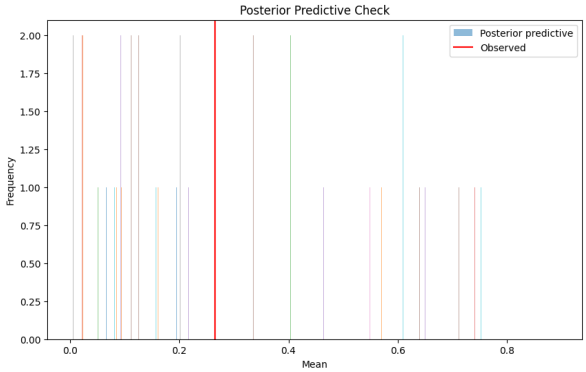



Figure 7: Posterior Predictive Check

Fiber optic service shows the highest odds of churn (OR = 2.72, CI: 0.59–7.92), with notable uncertainty. Total charges (OR = 1.86, CI: 1.34–2.53) and monthly charges (OR = 1.52, CI: 0.71–2.88) also increase churn likelihood. Senior citizen status (OR = 1.44), paperless billing (OR = 1.42), and multiple lines (OR = 1.27) exhibit moderate positive associations, while DSL service (OR = 1.21) shows uncertainty. Protective factors include phone service (OR = 0.42), tech support (OR = 0.53), online security (OR = 0.54), and tenure (OR = 0.19), the strongest factor, reducing odds by 81

### 4.6.1 Odds Ratio vs. Credible Interval

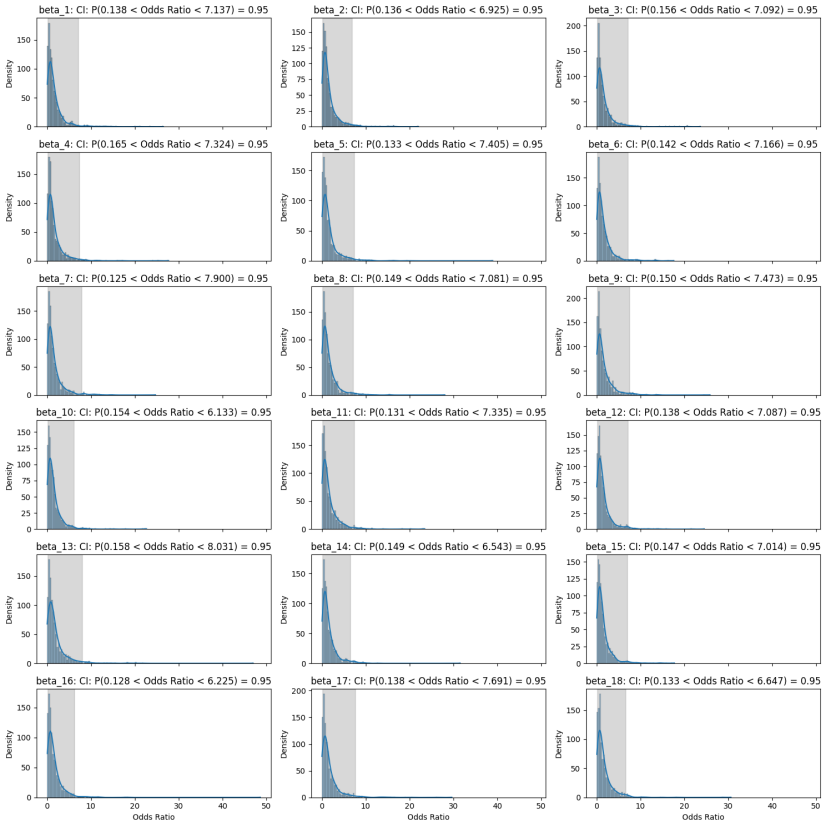



Figure 8: Odds Ratio vs. Credible Interval

The 95% credible intervals for all parameters have lower bounds ranging from approximately 0.107 to 0.166 and upper bounds from approximately 6.133 to 8.031. None of the credible intervals include zero, indicating statistically meaningful effects. The narrow credible intervals suggest precise parameter estimation. The right-skewed distributions reflect a tendency toward positive effects, confirming stable model estimation.

### 4.7 Model Reliability

The model demonstrates strong reliability across multiple metrics. All $\hat{R}$ values are equal to 1.0, indicating excellent chain convergence. Effective sample sizes for both bulk and tail estimates are high, ensuring reliable posterior estimates. Monte Carlo standard errors are consistently low, reflecting precise parameter estimation and model stability.

### 4.8 Results: Bayesian Approach

The Bayesian logistic regression model was evaluated using accuracy, precision, recall, F1 score, and ROC-AUC. The model showed strong performance, with minor performance decline on the test set, suggesting potential overfitting or sensitivity to unseen data.
**Model Evaluation Metrics**

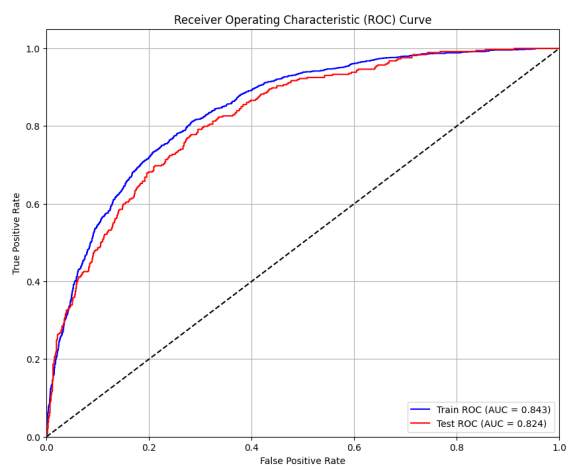
- **Training Set:** Accuracy 80.6%, Precision 66.7%, Recall 53.8%, F1 Score 59.6%, ROC-AUC 0.843.
- **Test Set:** Accuracy 78.8%, Precision 63.1%, Recall 48.9%, F1 Score 55.1%, ROC-AUC 0.824.

Despite lower recall, the model's high ROC-AUC values (above 0.8) for both datasets indicate effective prediction of churn and non-churn customers.

Model Evaluation Metrics:

| | Accuracy | Precision | Recall | F1 Score | ROC AUC |
|---|---|---|---|---|---|
| Train | 0.805867 | 0.667220 | 0.537793 | 0.595556 | 0.842860 |
| Test | 0.788202 | 0.631034 | 0.489305 | 0.551205 | 0.823955 |

(a) Performance: Bayesian Model




(b) ROC-AUC Curve

Figure 9: Bayesian Logistic Regression Performance on Test Set

The ROC-AUC curve (Figure 9b) shows clear separation between churn and non-churn classes, with consistent AUC values across both datasets.

## 5 Frequentist Approach

The frequentist logistic regression model also demonstrated strong performance, with minimal overfitting as reflected by similar metrics for training and test sets.
**Model Evaluation Metrics**

- **Training Set:** Accuracy 80.6%, Precision 65.96%, Recall 55.99%, F1 Score 60.6%, ROC-AUC 0.851.

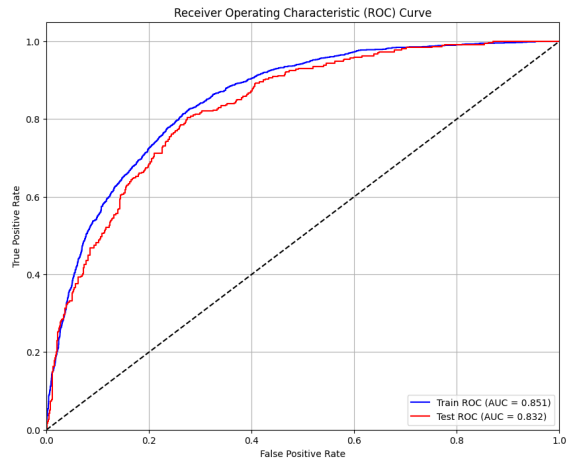
- **Test Set:** Accuracy 78.7%, Precision 61.94%, Recall 51.34%, F1 Score 56.14%, ROC-AUC 0.832.

The frequentist model displayed slightly better recall and F1 scores, with high ROC-AUC values (above 0.83) indicating strong performance in differentiating churn and non-churn customers.

Model Evaluation Metrics:

| | Accuracy | Precision | Recall | F1 Score | ROC AUC |
|---|---|---|---|---|---|
| Train | 0.806222 | 0.659574 | 0.559866 | 0.605644 | 0.851260 |
| Test | 0.786780 | 0.619355 | 0.513369 | 0.561404 | 0.831828 |

(a) Performance: Frequentist Model




(b) ROC-AUC Curve

Figure 10: Frequentist Logistic Regression Performance on Test Set

The ROC-AUC curve (Figure 10b) indicates strong discriminatory power, with AUC values of 0.851 for training and 0.832 for test datasets, confirming consistent performance.

# 6 Conclusion

This study compared Bayesian and frequentist logistic regression for predicting customer churn. Both models demonstrated similar performance across key metrics. Bayesian logistic regression achieved a training accuracy of 80.6% and test accuracy of 78.8%, with a test ROC-AUC of 0.824. The frequentist model showed comparable performance, with a training accuracy of 80.6%, test accuracy of 78.7%, and a test ROC-AUC of 0.832. Although the frequentist model slightly outperformed in recall and F1 score, the Bayesian approach offers distinct advantages.

Bayesian logistic regression excels in uncertainty quantification by providing credible intervals, offering deeper insights into parameter confidence. Its ability to incorporate prior information makes it robust in scenarios with limited or noisy data. Additionally, posterior distributions enhance interpretability, making model outputs more transparent and actionable for stakeholders. Despite its slightly lower recall, the Bayesian framework's probabilistic nature provides a powerful tool for decision-making, especially when uncertainty estimation is crucial.

# 7 Future Work

Future work could focus on incorporating more complex priors, such as informative or hierarchical priors, to enhance model performance. Addressing imbalanced data through Bayesian oversampling or cost-sensitive modeling could improve recall for underrepresented classes. Additionally, comparing Bayesian and frequentist logistic regression with non-linear models like Bayesian neural networks or random forests may reveal performance differences. Time-series analysis could be integrated to capture temporal patterns in churn, particularly for subscription-based services. Lastly, developing Bayesian online learning models for real-time predictions and continuous updates from new data could further enhance predictive accuracy and responsiveness.

# 8 References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